

Randomized and low-rank approximation

Improve the randomized LU squared-error factor to 2^k

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: Challenging because nonsymmetric residuals lose the trace identities behind Cholesky bounds; specialist impact is a sharper guarantee for one randomized LU rule.

Topic: randomized LU; low-rank approximation

Resolution — 2026-09-11

Negative resolution by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. Section 2 gives the exact counterexample $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: one pivot has expected squared Frobenius error $18/5$, while the best rank-one squared error is 1. Thus the displayed 2^k bound is false already at $k = 1$. Theorem 1 additionally proves that the known 4^r factor is sharp as a supremum at every rank, even on real entrywise-positive positive-definite inputs.

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Problem statement

For $A \in \mathbb{C}^{m \times n}$, define exact-arithmetic residuals $S_0 = A$. At step t , choose (i, j) with conditional probability $|(S_t)_{ij}|^2 / \|S_t\|_F^2$ and update

$$S_{t+1} = S_t - \frac{S_t(:, j)S_t(i, :)}{(S_t)_{ij}}.$$

After a zero residual, keep subsequent residuals zero. Prove or refute, for every $m, n \geq 1$, every A , and every $1 \leq k \leq \min(m, n)$,

$$\mathbb{E}\|S_k\|_F^2 \leq 2^k \sum_{j>k} \sigma_j(A)^2,$$

where singular values decrease with j . The bound concerns the mean squared Frobenius error of this specified algorithm.

References

Gilles and Wilber, *Low-Rank Approximation by Randomly Pivoted LU*, Algorithm 1 and §3.1, Theorem 3, Eq. (10), and the following conjecture (pp. 7–8).

Status check

Their theorem gives 4^k ; the conjecture explicitly replaces it by 2^k . Searches for the title, *randomly pivoted LU* 2^k , and improved RPLU error bounds found no later resolution. The arXiv submission history listed only v1 of January 29, 2026. The August 2026 RPCholesky theorem addresses a different pivot distribution and norm.

Audit — 2026-09-10

Rechecked [Algorithm 1](#), [Theorem 3](#), and [the following conjecture](#); its record still lists v1. RPLU and improved-bound searches found no resolution. The theorem remains at 4^k , while the stated squared-error conjecture is 2^k .